Unit 6, Lesson 21

# Problems with Remainders

## Standards

Building On **4.NBT.B.5, 4.OA.B.4**

Addressing **4.NBT.B.6, 4.OA.A.3**

## Instructional Routines

- Number Talk

## Goals

- Explain (orally and in writing) strategies for determining whether a number is a multiple or factor of another.

## Student Facing Learning Goals

Let's think about remainders to solve problems.

## Lesson Purpose

The purpose of this lesson is for students to apply their understanding of division, multiplication, factors, and multiples to solve problems involving remainders.

## Narrative

In this lesson, students encounter problems about finding unknown factors and relate such problems to division. The context in this lesson allows students to apply their understanding of factors and multiples developed in an earlier unit. The dividends in this lesson are 3-digit numbers beyond 100.

Students continue to interpret division in terms of finding the number of groups ("If we write multiples of 5, how many numbers will we need to write to get to 105?") and the size of a group ("What number are we finding multiples of if we get to 112 after writing 7 numbers?"). Students may solve the problems by multiplying in parts (finding partial products) or by dividing in parts (finding partial quotients). Through repeated reasoning, they notice that it helps to decompose a dividend into familiar multiples (MP2).

In these materials, division that results in a whole number with a remainder—for example $145 \div 7$—is not expressed with an expression such as "20 R 5." Instead, students will relate this result to a multiplication equation, $145 = 7 \times 20 + 5$.

## Access for Students with Disabilities

- Engagement

## Access for English Learners

- MLR8

## Required Materials

**Materials to Gather**

- Number Cards 0–10: Activity 3
- Paper clips: Activity 3

## Lesson Timeline

| | |
|---|---|
| Warm-up | 10 min |
| Activity 1 | 20 min |
| Activity 2 | 15 min |
| Activity 3 | 15 min |
| Synthesis Estimate | 10 min |
| Cool-down | 5 min |

## Teacher Reflection Questions

Which questions that you asked today would you rephrase to improve students' ability to make connections or to help them better consolidate what they did? How would you rephrase them?

# Warm-up

10 min

Number Talk: Dividing by 7

## Standards

Addressing **4.NBT.B.6**

## Instructional Routines

- Number Talk

This *Number Talk* encourages students to compose or decompose multiples of 7 and to rely on properties of operations to mentally solve problems. The ability to compose and decompose numbers will be helpful when students divide multi-digit numbers. It also promotes the reasoning that is useful when finding multiples of a number, or when deciding if a number is a multiple of another number.

## Student Task Statement

Find the value of each expression mentally.

- $21 \div 7$
- $35 \div 7$
- $140 \div 7$
- $196 \div 7$

## Student Response

- 3: Three times 7 is 21.
- 5: Five times 7 is 35.
- 20: Two times 7 is 14, so 20 times 7 is 140.
- 28: Sample responses:
  - $196 = 140 + 56$, and $56 = 21 + 35$. This means $196 = 140 + 21 + 35$, so $196 = (20 \times 7) + (3 \times 7) + (5 \times 7)$.
  - Three times 7 is 21, so 30 times 7 is 210. 196 is 14 less than 210, which means it is $2 \times 7$ less than $30 \times 7$.

## Launch

- Display one expression.
- "Give me a signal when you have an answer and can explain how you got it."
- 1 minute: quiet think time

## Activity

- Record answers and strategy.
- Keep expressions and work displayed.
- Repeat with each expression.

## Activity Synthesis

- "What do the expressions have in common?" (They all involve division by 7. The dividends are all multiples of 7. The results have no remainders.)
- "How did the first three expressions help us find the value of the last expression?"
- Consider asking:

- "Who can restate _____ 's reasoning in a different way?"
- "Did anyone have the same strategy but would explain it differently?"
- "Did anyone approach the expression in a different way?"
- "Does anyone want to add on to___'s strategy?"

# Activity 1

20 min

Write Multiples

## Standards

| | |
|---|---|
| Building On | **4.NBT.B.5, 4.OA.B.4** |
| Addressing | **4.NBT.B.6** |

The purpose of this activity is to prompt students to use the relationship between multiplication and division and their understanding of factors and multiples to solve problems about an unknown factor (MP7). Students recognize such problems as division situations. In this activity, the dividends are three-digit numbers and the divisors are one-digit numbers. Students use the context of factors and multiples to interpret division that results in a remainder.

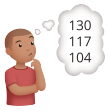
One approach for solving these problems is by decomposing the dividend into familiar multiples of 10 and then dividing the remaining number (which is now smaller) by the divisor. Another is to find increasingly greater multiples of the divisor until reaching the dividend. Both are productive and appropriate. In the *Activity Synthesis*, help students see the connections between the two paths.

## Access for Students with Disabilities

*Engagement: Develop Effort and Persistence.* If students don't recognize this as a division situation at first, they may do so when presented with a more accessible value. Consider offering this situation first: "Han starts writing multiples of a number. When he reaches 12, he has written 3 numbers." Invite students to identify what number Han is writing multiples of (4), and how they know. Then ask how they might apply that reasoning to the task as presented here.
*Supports accessibility for: Conceptual Processing*

## Student Task Statement

1. Han starts writing multiples of a number. When he reaches 104, he has written 8 numbers.




   For each of the following questions, explain or show your reasoning.

   a. What number is Han writing multiples of?

   b. What is the 15th multiple of this number?

## Launch

- "Who can remind the class of the meaning of 'multiple?'"

## Activity

- 4–5 minutes: independent work time for the first set of questions
- Monitor for students who:

c. Han gets to 286. How many numbers has he written at that point?

2. Kiran wants to know how many multiples of 7 are between 0 and 150.
   a. He thinks he can use division to find out. Do you agree? Explain your reasoning.
   b. How many multiples will he find? Show your reasoning.
   c. Is 150 a multiple of 7? Show how you know.
3. Han writes the number 295. He is wondering if he made a mistake. Determine if 295 is a multiple of 8 and explain your reasoning.

## Student Response

1. a. 13. Sample responses:
   - $8 \times 10 = 80$ and $80 + 24 = 104$. Since 24 is $8 \times 3$, Han must be finding multiples of $10 + 3$ or 13.
   - $80 \div 8 = 10$ and $24 \div 8 = 3$, so $104 \div 8 = 13$.

   b. 195, because $15 \times 13 = 195$.

   c. 22 numbers. Sample responses:
   - $(10 \times 13) + (10 \times 13) + (2 \times 13) = 286$
   - $20 \times 13 = 260$ and $2 \times 13 = 26$, so $22 \times 13$ is $260 + 26$, which is 286.

2. a. Agree. Sample response: He can divide 150 by 7 because he's trying to find how many 7s are in 150.

   b. 21. Sample responses:
   - There are twenty 7s in 140, plus one more 7 makes 147. This is as high as we could get without going over 150.
   - $20 \times 7 = 140$ and $1 \times 7 = 7$, so $21 \times 7 = 147$, which is less than 150.

   c. No. $7 \times 21 = 147$ and $7 \times 22 = 154$

3. No. 295 is not a multiple of 8 because $286 + 8 = 294$.

- Decompose 104 into a multiple of 8 and another number.
- Compose 104 from increasingly larger multiples of 8.
- Use multiplication or division equations to show their reasoning.

- Pause for a discussion before the second set of questions.
- Select students who use different decomposition strategies to share responses. Record and display their reasoning.
- "Let's revisit each question we just answered. What equations could we write to represent them?"
- Display equations and highlight their connection to the questions:
  a. $8 \times ? = 104$ or $104 \div 8 = ?$
  b. $15 \times 13 = ?$ or $13 \times 15 = ?$
  c. $? \times 13 = 286$ or $286 \div 13 = ?$
- 4–5 minutes: independent work time for the second set of questions

## Activity Synthesis

- Invite students to share responses for the last set of questions.
- Highlight that to divide a number by a smaller number—say, divide 150 by 7, we can:
  - Use familiar multiples or multiplication facts to help us. For example, if we know $(20 \times 7) + (1 \times 7) = 147$, we know the result is 21 with a remainder of 3, or $150 = 21 \times 7 + 3$.
  - Think of the dividend in smaller chunks. For example: We can see the 150 as $140 + 10$ and divide each 140 and 10 by 7 separately, which gives $20 + 1$ with a remainder of 3.

# Activity 2

15 min

Jada's Mystery Number

## Standards

Addressing **4.NBT.B.6**

The purpose of this optional activity is for students to continue to use the relationship between multiplication and division to reason about situations that involve division. Students divide three-digit numbers by single-digit divisors and find results with and without a remainder. This activity provides more practice applying the relationship between multiplication and division.

*MLR8 Discussion Supports.* Before they begin, invite students to make sense of the situation. Monitor and clarify any questions about the context.
*Advances: Reading, Representing.*

## Student Task Statement

Jada writes multiples of a mystery number. After writing some numbers, she writes the number 126.

- Mai says 6 is the mystery number.
- Priya says 8 is the mystery number.
- Andre says 9 could be the mystery number.

1. Which student do you agree with? Show how you know using equations.
2. Jada gives one more clue: "If I keep writing multiples, I'll get to 153."

   What is the mystery number? Explain or show your reasoning.

## Student Response

1. Sample responses:
   - I agree with Mai, because 126 is $120 + 6$, and 120 is a multiple of 6 ($6 \times 20 = 120$) and 6 is the first multiple of 6 ($6 \times 1 = 6$).
   - I agree with Andre and Mai because 126 is a multiple of both 9 and 6, because when counting by each number I say 126.
2. 9. Sample response: 153 and 126 are both multiples of 9.

## Launch

- Groups of 2

## Activity

- 5 minutes: independent work time on the first question
- 2 minutes: partner discussion
- 2–3 minutes: partner work time on the second question
- Monitor for students who clearly show how they use partial products or partial quotients to answer the questions.

## Activity Synthesis

- "How could we use division to help us find the mystery number?" (If the result has a remainder, then the divisor could not be the mystery number. 153 divided by 6 has 3 as a remainder. 153 divided by 9 has no remainder, so 9 is the mystery number.)

# Activity 3

15 min

Watch Your Remainder!

## Standards

Addressing **4.NBT.B.6**

The purpose of this activity is to introduce the center activity *Watch Your Remainder*, Stage 1. Students spin a spinner to get the divisor for the round. Each student picks 5 cards and chooses 3–4 of them to create a dividend. Each student finds their quotient. The score for the round is the remainder from each expression. Students pick new cards so they have 6 cards in their hand and then start the next round. The player with the lowest score after 6 rounds wins.

The *Activity Synthesis* provides suggested cards to display to discuss strategies for the game. Consider having a group of 2 share one of the rounds from their game instead.

## Required Materials

**Materials to Gather**

- Number Cards 0–10: Activity 3
- Paper clips: Activity 3

## Required Preparation

Remove (or have students remove) the cards that show 10 from the Number Card deck.

## Student Task Statement

Directions:

- Spin the spinner to get your one-digit divisor.
- Each partner:
  - Use 3–4 cards to create a dividend.
  - Write a multiplication equation to represent the quotient. (For example, $109 \div 9$ is written as $(9 \times 12) + 1$ and your score is 1.)
  - Check your partner's work to make sure you agree.
  - Your score for each round is the remainder.
- Take new cards so that you have 4 cards to start the next round.
- The partner who has the fewest points once the recording sheet is full wins the game.

## Launch

- Groups of 2
- Give each group a set of Number Cards and 2 copies of the recording sheet.
- "We are going to learn a game called *Watch Your Remainder*."
- Give students several minutes to read over the directions.
- Answer any clarifying questions from students.
- Consider playing a sample round with the class.

## Activity

- 10 minutes: partner work time
- Monitor for students who play a round where there is no remainder and one where there is a remainder.

## Activity Synthesis

- Invite selected students to show the number cards

### Student Response

Answers vary.

for the round with no remainder or display the number cards 0, 1, 2, 3, 7 with a divisor of 5.

- "How would you use these cards to get the smallest remainder?"
- Invite selected students to show the number cards for the round with a remainder or display the number cards 0, 1, 2, 3, 7 with a divisor of 9.
- "How would you use these cards to get the smallest remainder?"

## Lesson Synthesis

"Today we tried to find out if a number is a multiple or a factor of another number. For example: Is 267 a multiple of 8?"

"Is this question a division problem?" (Yes)

"Why?" (We're looking for how many 8s are in 267.)

"What are we dividing?" (267 by 8)

"One way to answer the question is by using familiar multiplication facts or by finding partial products. How would you start?" (One way is to start with 10 x 8 or its multiples, build the products up to 267 or close to it, and then try smaller multiples of 8.)

"Why might it be helpful to start with multiples of 10?" (They're easy to find and easy to add.)

"Can we use division to answer the question?" (We can start with a number close to 267 that is a multiple of 8, divide it by 8, see what is left, and find a multiple of 8 that is close to that number. For example: $160 \div 8 = 20$. After taking 160 away, there's 107 left. $80 \div 8 = 10$. After taking 80 away, there's 27 left. $24 \div 8 = 3$. After taking 24 away, there's 3 left, which is not enough to make 8.)

"How are the two approaches alike?" (They involve using smaller multiples of a number to see if a larger number is a multiple of that number.)

### Suggested Centers

- Compare (1–5), Stage 4: Divide within 100 with One-Digit Divisors (Supporting)
- Rolling for Fractions (3–5), Stage 2: Multiply a Fraction by a Whole Number (Supporting)

## Cool-down

5 min

Reaching 161 with Multiples

### Standards

Addressing **4.NBT.B.6, 4.OA.A.3**

## Student Task Statement

Mai is writing multiples of 7. Mai wonders whether she will say the number 149. Do you think she will? Explain or show your reasoning.

## Student Response

She will not. Sample response: 140 is a multiple of 7 because $7 \times 2 = 14$. So, $7 \times 20 = 140$. 149 is the same as $140 + 9$, so there would be a remainder of 2.

## Responding to Student Thinking

| | |
|---|---|
| Students do not write a division equation to represent the situation. | Next Day Supports<br>Launch the *Warm-up* or first activity by highlighting important notation from previous lessons. |
| The work in this lesson builds from the concept of factors and multiples developed in a prior unit. | Prior-Unit Support<br>Grade 4, Unit 1, Section A Understand Factors and Multiples |

## Section C Summary

We solved different problems that involved dividing whole numbers.

We recalled two ways of thinking about division.

For example, if $274 \div 8$ represents a situation where 274 markers are put into equal groups. The value of $274 \div 8$ can tell us:

- How many markers are in each group if there were 8 groups.
- How many groups can be made if there were 8 markers in each group.

We learned that in $274 \div 8$, the $274$ is called the **dividend**, and the $8$ is called the **divisor.** We then identified many ways to find the value of a **quotient** —or the result of the division. For $274 \div 8$, we can:

- Think about whether one number is a multiple or factor of another number. For example, "Is 274 a multiple of 8?" or "Is 8 a factor of 274?"
- Divide by place value and think about putting 2 hundred, 7 tens, and 4 ones into 8 equal groups.
- Divide in parts and find partial quotients. For example, we can first find $160 \div 8$ (which is 20), and then $80 \div 8$ (which is 10), and then $32 \div 8$ (which is 4).
- Think in terms of multiplication. For example, we can think of $8 \times 20 = 160$, $8 \times 10 = 80$, and so on.

Here is one way to record division using partial quotients.

$$
\begin{array}{rl}
\boxed{34} & \\
4 & \\
10 & \\
20 & \\
8\overline{)274} & \\
-\ 160 & 8 \times 20 \\
\hline
114 & \\
-\ 80 & 8 \times 10 \\
\hline
34 & \\
-\ 32 & 8 \times 4 \\
\hline
2 &
\end{array}
$$

Sometimes a division results in a leftover that can't be put into equal groups or is not enough to make a new group. We call the leftover a **remainder**. Dividing 274 by 8 gives 34 and a remainder of 2.